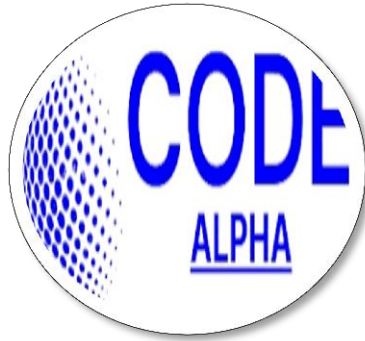


CODE ALPHA
Empowering Future Innovators



INTERNSHIP REPORT
On
BIOINFORMATICS

Task 2: MULTIPLE SEQUENCE ALIGNMENT

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Submitted to: Code Alpha

MULTIPLE SEQUENCE ALIGNMENT

Cross-Species Comparison of the Insulin Protein Family

CodeAlpha Bioinformatics Internship — Task 2

1. Introduction

Multiple Sequence Alignment (MSA) is a fundamental technique in bioinformatics used to arrange three or more biological sequences so that regions of similarity can be identified and compared side by side. Unlike a pairwise alignment, which compares only two sequences, an MSA reveals patterns of conservation and divergence across an entire protein family, making it possible to infer evolutionary relationships and pinpoint functionally important regions of a protein.

This report applies MSA to the insulin protein family, using five sequences drawn from different vertebrate species. The goal is to identify which regions of the insulin sequence are conserved across species and therefore likely essential to its biological function and which regions show greater variability, reflecting evolutionary divergence.

2. Methodology

The following workflow was used to generate the alignment:

- Five reviewed (Swiss-Prot) insulin sequences were retrieved from UniProt (uniprot.org), each in FASTA format, covering a mix of mammalian and avian species (Table 1).
- The five FASTA sequences were combined into a single input file, with each sequence header preserved.
- The combined file was submitted to Clustal Omega (EMBL-EBI, ebi.ac.uk/jdispatcher/msa/clustalo) using default alignment parameters.
- The resulting alignment was examined in both plain-text and colour-coded formats, and the accompanying guide tree was used to view the inferred evolutionary relationships between the five sequences.

Table 1. The five insulin sequences used in this alignment.

UniProtKB 5,400 results

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Entry	Entry Name	Protein Names	Gene Names	Organism	Length
☒ P01308	INS_HUMAN	Insulin[...]	INS	Homo sapiens (Human)	110 AA
☐ P06213	INSR_HUMAN	Insulin receptor[...]	INSR	Homo sapiens (Human)	1,382 AA
☐ P14735	IDE_HUMAN	Insulin-degrading enzyme[...]	IDE	Homo sapiens (Human)	1,019 AA
☒ P01317	INS_BOVIN	Insulin[...]	INS	Bos taurus (Bovine)	105 AA
☒ P01321	INS_CANLF	Insulin[...]	INS	Canis lupus familiaris (Dog) (Canis familiaris)	110 AA
☒ P01315	INS_PIG	Insulin[...]	INS	Sus scrofa (Pig)	108 AA
☒ P67970	INS_CHICK	Insulin[...]	INS	Gallus gallus (Chicken)	107 AA
☐ Q91X13	INS_ICTTR	Insulin[...]	INS	Ictidomys tridecemlineatus (Thirteen-lined ground squirrel) (Spermophilus tridecemlineatus)	110 AA
☐ P01329	INS_CAVPO	Insulin[...]	INS	Cavia porcellus (Guinea pig)	110 AA
☐ P17715	INS_OCTDE	Insulin[...]	INS	Octodon degus (Degu) (Sciurus degus)	109 AA

Mammals (human, cattle, dog, pig) were combined with one bird species (chicken) deliberately, so that the alignment would show conservation both within closely related mammals and across a wider evolutionary gap between mammals and birds.

3. Results

Clustal Omega successfully aligned all five sequences. Figure 1 shows the raw text alignment output, and Figure 2 shows the same alignment in a colour-coded viewer, where each amino acid is shaded by chemical property (Clustal2 colour scheme).

Tool Output Alignments Guide Tree Phylogenetic Tree Results Viewers Result

Tool output

Download

CLUSTAL O(1.2.4) multiple sequence alignment

```

sp|P67970|INS_CHICK      MALWIRSLPLLALLVFSGPGTSYAAANQHLCGSHLVEALYLVCGERGFFYTPKARRDVEQ  60
sp|P01317|INS_BOVIN     MALWTRLRPLLALLALWPPPPARAFVNQHLGSHLVEALYLVCGERGFFYTPKARREVEG  60
sp|P01308|INS_HUMAN     MALWMRLPLLALLALWGPDPAAAFVNQHLGSHLVEALYLVCGERGFFYTPKTRREAED  60
sp|P01321|INS_CANLF     MALWMRLPLLALLALWAPAPTRAFAVNQHLGSHLVEALYLVCGERGFFYTPKARREVED  60
sp|P01315|INS_PIG       MALWTRLPLLALLALWAPAPAQAFVNQHLGSHLVEALYLVCGERGFFYTPKARREAEN  60
*****
PLVSSPL-RGE---AGVLPFQEEYEKVKRGIVEQCCHNTCSLYQLENYCN  107
PQVGALEAGGPGAGGL-----EGPPQKRGIVEQCCASVCSLYQLENYCN  105
LQVGQVELGGGPGAGSLQPL-ALEGSLQKRGIVEQCCTISCSLYQLENYCN  110
LQVRDVELAGAPGEGGLQPL-ALEGALQKRGIVEQCCTISCSLYQLENYCN  110
PQAGAVELGGG--LGGLQAL-ALEGPPQKRGIVEQCCTISCSLYQLENYCN  108
*****

```

Coloured sequences

Show

Figure 1. Clustal Omega text alignment output for the five insulin sequences. Rows of symbols beneath each block mark conservation: '*' = fully conserved, ':' = strongly similar, '.' = weakly similar, blank = variable.

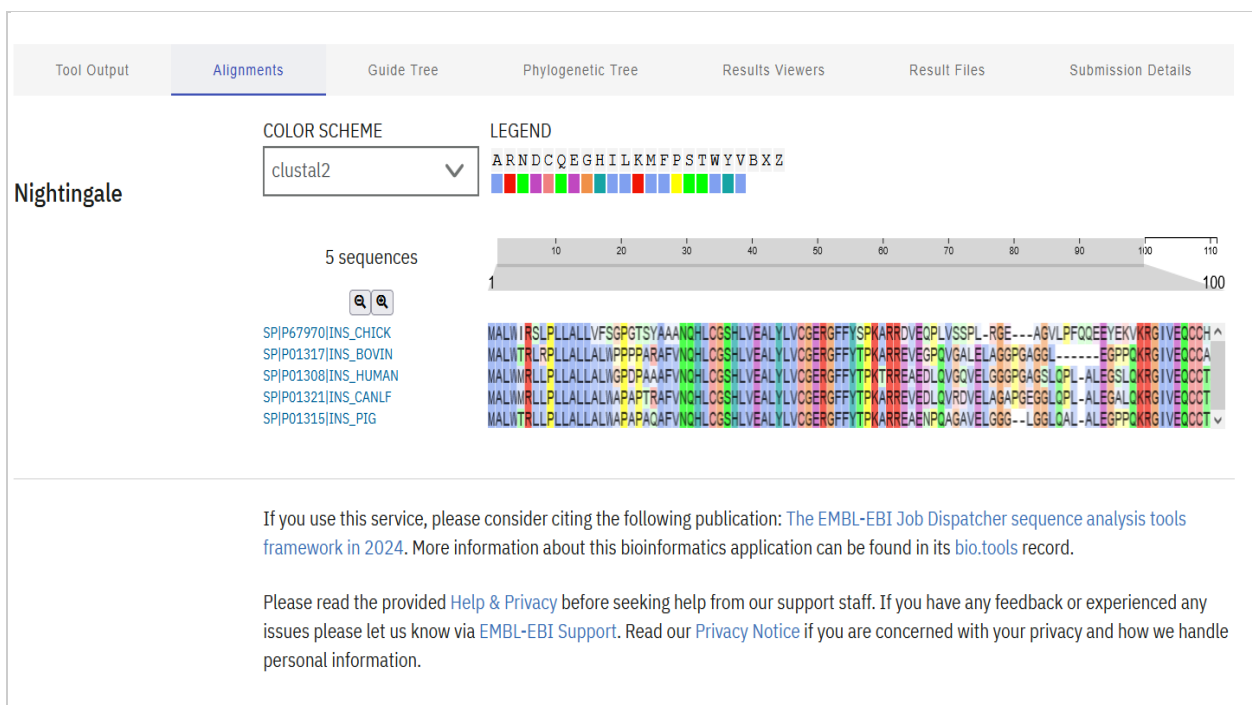


Figure 2. Colour-coded alignment view (Clustal2 scheme) showing the same five sequences, with amino acids shaded by chemical property to make conserved blocks visually easier to spot.

The alignment shows a long, highly conserved stretch at the start of the sequence — largely unbroken rows of asterisks — corresponding to the signal peptide and the B-chain region of insulin. In contrast, the region joining the B-chain and A-chain (roughly positions 60–107) shows far more variation, including outright gaps: *Bos taurus* (cattle) and *Gallus gallus* (chicken) both show a run of dashes where other species retain extra residues, indicating an insertion/deletion (indel) event during evolution.

Figure 3 shows the guide tree generated alongside the alignment, which groups the five sequences by similarity.



Figure 3. Phylogenetic guide tree and Newick-format branch data for the five insulin sequences, showing branch lengths as a measure of evolutionary distance.

4. Interpretation

- **Conserved regions:** The first ~35 residues of the alignment (signal peptide and start of the B-chain, including the motif ...NQHLCGSHLVEALYLVCGERGFFY...) are almost perfectly conserved across all five species, marked by continuous asterisks. This region is critical to insulin's structure and receptor-binding function, so mutations here are strongly selected against.
- **Variable / gapped region:** The C-peptide-like linker region (roughly positions 60–100) shows the weakest conservation and contains gaps in the cattle and chicken sequences. This part of the precursor is cleaved out during insulin processing and does not appear in the mature, active hormone, which explains why it can tolerate far more mutation without harming function.
- **A-chain conservation:** Toward the end of the alignment, conservation rises again (dense asterisks around ...GIVEQCC...SICSLYQLENYCN), corresponding to the A-chain, which — together with the B-chain — forms the mature, functional insulin molecule.
- **Phylogenetic grouping:** The guide tree places cattle closest to chicken and groups human with dog, with pig branching separately — driven purely by sequence similarity rather than

the accepted mammalian evolutionary tree. Branch lengths (e.g. 0.26272 for chicken vs. ~0.05–0.06 for the mammals) show chicken is clearly the most diverged sequence, consistent with birds and mammals having split much earlier than the mammalian species diverged from one another.

Overall, the alignment demonstrates that functionally essential regions of insulin — the B-chain and A-chain that make up the active hormone — are strongly conserved across a wide evolutionary span, while the non-functional linker region has drifted more freely. This is a textbook example of how sequence conservation tracks functional importance.

5. Conclusion

This exercise used Clustal Omega to perform a multiple sequence alignment of five insulin orthologs spanning four mammalian species and one bird species. The alignment identified clearly conserved blocks corresponding to the functional B-chain and A-chain of insulin, alongside a more variable, gapped linker region that is removed during hormone maturation. These findings reinforce the principle that regions under strong functional constraint are conserved through evolution, while non-essential regions accumulate mutations and indels more freely, and illustrate how MSA can be used to connect sequence-level patterns to biological function.

References

UniProt Consortium. UniProt Knowledgebase (UniProtKB). Available at: <https://www.uniprot.org/>

EMBL-EBI. Clustal Omega Multiple Sequence Alignment Tool. Available at: <https://www.ebi.ac.uk/jdispatcher/msa/clustalo>